

Martin Pawelczyk

My research focuses on developing **rigorous empirical and theoretical frameworks for advancing AI Safety**, specifically through **privacy, unlearning, robustness and interpretability**. During my PhD, I uncovered fundamental tradeoffs among these core properties, for instance, how improving interpretability can inadvertently compromise privacy and robustness. My work also challenged the prevailing assumption that privacy must come at the cost of utility, showing instead how strategic design can optimize both. Along the way, I led the development of several open-source software packages, now utilized by hundreds of users across research and industry. The next generation of safe AI algorithms must ensure that pretrained large language models (LLMs) adapt reliably to specific safety objectives while maximizing data efficiency. Towards this goal, my recent work introduces fundamentally novel ways of unlearning data from LLMs to protect end-user privacy. Through these contributions, I aim to advance the practical and theoretical frontiers of AI in building systems that stakeholders can reliably trust.

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Research Interests

AI Trustworthiness, Data Centric AI
Large Language Models, Agents
Private training, scaling, inference
Evaluation, Open Source Software

Positions

- '26 Feb – Present ♦ **Assistant Professor for Responsible Computing, University of Vienna.**
- '23 July – '25 Oct ♦ **Postdoctoral Fellow, Harvard University.**
Hosts: [Himabindu Lakkaraju](#) and [Seth Neel](#).
LLM privacy through the lens of unlearning.
- '21 Apr – '23 June ♦ **Research Fellow, Harvard University.**
Hosts: [Himabindu Lakkaraju](#).
Explainable & private machine learning methods.
- '21 Nov – '22 May ♦ **Research Fellow, Max-Planck-Institute for Security & Privacy, Bochum**
Host: [Asia Biega](#)
Explainable machine learning methods in the context of GDPR's right to be forgotten.
- '21 May – '21 Aug ♦ **AI Research Intern, JP Morgan, London**
Host: [Daniele Magazzini](#)

Education

- '18 – '23 ♦ **Ph.D. Computer Science, University of Tübingen, Germany.**
On the Generation of Realistic and Robust Counterfactual Explanations for Algorithmic Recourse
Advisor: [Gjergji Kasneci](#), Reviewer: [Ulrike von Luxburg](#).
- '17 – '18 ♦ **M.Sc. Statistics (Research), London School of Economics (LSE), UK.**
With high distinction for research thesis.
- '16 – '17 ♦ **M.Sc. Econometrics, University of Edinburgh, UK.**
With distinction.
- '13 – '16 ♦ **B.Sc. Economics, University of Cologne, Germany.**
Ranked top 3% out of 351 students.

Selected Honors and Awards

- 2025 ♦ **Appointment as Assistant Professor, TU Darmstadt/Fraunhofer, NTU Singapore & UCL**
(all declined)
- 2024 ♦ **Spotlight Paper** 🏆, AAAI'24
- 2021, 2023, 2024 ♦ **Spotlight Presentation** 🏆, ICML'21 WS, ICLR'22 WS, ICML'24 WS
- 2022, 2024 ♦ **Top Reviewer Award** 🏆, NeurIPS'22, NeurIPS'24
- 2023 ♦ **Travel Scholarship, NeurIPS'23, ICLR'23, ICLR'25**
- 2016–2018 ♦ **German National Merit Foundation Scholarship, 40.000 Euros**
- 2016–2017 ♦ **University of Edinburgh Scholarship, 10.000 Euros**

Research Publications

H-index: 16, Citations: 3871

1. Abedini, S., Mavali, S., Schönherr, L., **Pawelczyk***, M., & Burkholz*, R. (2026). **Don't Trust Stubborn Neighbors: A Security Framework for Agentic Networks**. *arXiv preprint: arXiv:2603.15809*.
2. Bordt, S., & **Pawelczyk**, M. (2026). **Train Once, Answer All: Many Pretraining Experiments for the Cost of One**. In *International Conference on Learning Representations (ICLR)*.
3. Hanke, V., Xu, J., **Pawelczyk**, M., Backes, M., Dziedzic, A., & Boenisch, F. (2026). **Learn to be Unlearned: Optimizing Language Models for Unlearning via Clustered Gradient Routing**. *in submission*.
4. Karchmer, A., Neel, S., & **Pawelczyk**, M. (2025). **Efficiently Verifiable Proofs of Data Attribution**. In *Advances in Neural Information Processing Systems (NeurIPS)*.
5. **Pawelczyk**, M., Di, J. Z., Lu, Y., Kamath, G., Sekhari, A., & Neel, S. (2025). **Machine Unlearning Fails to Remove Data Poisoning Attacks[†]**. In *International Conference on Learning Representations (ICLR)*, Spotlight Presentation 🏆 at ICML'24 GenLaw Workshop.
6. **Pawelczyk**, M., Sun, L., Qi, Z., Kumar, A., & Lakkaraju, H. (2025). **Generalizing Trust: Weak-to-Strong Trustworthiness in Language Models[†]**, Spotlight Presentations 🏆 at ICML'25 MOSS & CFAgentic Workshops. *arXiv preprint arXiv:2501.00418*.
7. Leemann, T., **Pawelczyk**, M., Eberle, C. T., & Kasneci, G. (2024). **I Prefer Not To Say: Protecting User Consent in Models with Optional Personal Data**. In *38th Annual AAAI Conference on Artificial Intelligence (AAAI)*, Spotlight Presentation 🏆.
8. **Pawelczyk**, M., Neel, S., & Lakkaraju, H. (2024). **In-Context Unlearning: Language Models as Few Shot Unlearners**. In *International Conference on Machine Learning (ICML)*.
9. Leemann*, T., **Pawelczyk***, M., & Kasneci, G. (2023). **Gaussian Membership Inference Privacy**. In *Advances in Neural Information Processing Systems (NeurIPS)*.
10. **Pawelczyk**, M., Lakkaraju*, H., & Neel*, S. (2023). **On the Privacy Risks of Algorithmic Recourse**. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*.
11. **Pawelczyk**, M., Leemann, T., Biega*, A., & Kasneci*, G. (2023). **On the Trade-Off between Actionable Explanations and the Right to be Forgotten**. In *International Conference on Learning Representations (ICLR)*.
12. **Pawelczyk**, M., Datta, T., van-den-Heuvel, J., Kasneci, G., & Lakkaraju, H. (2023). **Probabilistically Robust Recourse: Navigating the Tradeoffs between Costs and Robustness in Algorithmic Recourse[†]**. In *International Conference on Learning Representations (ICLR)*.
13. V. Borisov and K. Seßler and T. Leemann and M. **Pawelczyk** and G. Kasneci. (2023). **Large Language Models are Realistic Tabular Data Generators[†]**. In *International Conference on Learning Representations (ICLR)*.
14. Agarwal, C., Saxena, E., Krishna, S., **Pawelczyk**, M., Johnson, N., Puri, I., ... Lakkaraju, H. (2022). **OpenXAI: Towards a Transparent Evaluation of Model Explanations**. In *Advances in Neural Information Processing Systems (NeurIPS)*.
15. Borisov, V., Leemann, T., Seßler, K., Haug, J., **Pawelczyk**, M., & Kasneci, G. (2022). **Deep Neural Networks and Tabular Data: A Survey[†]**. In *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*.
16. **Pawelczyk**, M., Agarwal, C., Joshi, S., Upadhyay, S., & Lakkaraju, H. (2022). **Exploring Counterfactual Explanations Through the Lens of Adversarial Examples: A Theoretical and Empirical Analysis**. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*.
17. **Pawelczyk**, M., Bielawski, S., Van den Heuvel, J., Richter*, T., & Kasneci*, G. (2021). **CARLA: A Python Library to Benchmark Algorithmic Recourse and Counterfactual Explanation Algorithms[†]**. In *Advances in Neural Information Processing Systems (NeurIPS)*, Spotlight Presentation 🏆 at ICML'21 Algorithmic Recourse Workshop.

18. Haug, J., **Pawelczyk**, M., Broelemann, K., & Kasneci, G. (2020). **Leveraging model inherent variable importance for stable online feature selection**. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*.
19. **Pawelczyk**, M., Broelemann, K., & Kasneci, G. (2020). **On Counterfactual Explanations under Predictive Multiplicity**. In *36th Conference on Uncertainty in Artificial Intelligence (UAI)*.
20. **Pawelczyk**, M., Broelemann, K., & Kasneci, G. (2020). **Learning Model-Agnostic Counterfactual Explanations for Tabular Data**. In *Proceedings of The Web Conference 2020 (WWW)*.

Grants

- ◇ Co-Investigator of the grant: “Egocentric perception, interaction and computing in the deep learning era”
German Academic Exchange Service (DAAD)
Funding Amount: 80.000€
2020 – 2022

Open Source Software Projects

- 2026 ◇ TOAA: Enabling academia to contribute to pretraining research
<https://github.com/sbordt/pretrain-experiments/>
- 2023 ◇ GReaT: Generation of Realistic Tabular data with LLMs to enable private data sharing
https://github.com/kathrinse/be_great, **350+ github stars**
- 2022 ◇ OpenXAI: A library to evaluate feature attribution algorithms
<https://github.com/AI4LIFE-GROUP/OpenXAI>, **250+ github stars**
- 2021 ◇ CARLA: A library to evaluate counterfactual explanation algorithms
<https://github.com/carla-recourse/CARLA>, **300+ github stars**

Presentations

- 2026 ◇ *Beyond Retraining: Efficient Algorithms and Evaluations for Private Data Unlearning*,
Huawei Aurora Tech Summit, Keynote Talk, Helsinki
- 2025 ◇ *Verifying Data Attributions without Breaking the Bank*,
Winter School on Causality and Explainable AI, Sorbonne University, Paris
- ◇ *Building AI Safeguards: Computational Methods for Privacy Compliance*,
Technical University of Munich (TUM), Heilbronn
- ◇ *Building AI Safeguards: Computational Methods for Privacy Compliance*,
University of Vienna, Vienna
- ◇ *Privacy-Utility Tradeoffs in LLM Systems: From Data Selection to Agent Behavior*,
University College London (UCL), London
- ◇ *Building AI Safeguards: Computational Methods for Privacy & Explainability Compliance*,
Nanyang Technological University (NTU), Singapore
- ◇ *Building AI Safeguards: Computational Methods for Privacy & Explainability Compliance*,
TU Darmstadt, Darmstadt
- ◇ *Building AI Safeguards: Computational Methods for Privacy & Explainability Compliance*,
ETH Zurich, Zurich
- ◇ *Building AI Safeguards: Computational Methods for Privacy & Explainability Compliance*,
CISPA Helmholtz Center for Information Security, Saarbrücken
- ◇ *Building AI Safeguards: Computational Methods for Privacy & Explainability Compliance*,
Responsible Data Science Lab, hosted by Prof. Gjergji Kasneci, TU Munich (TUM)

Presentations (continued)

- ◇ *Building AI Safeguards: Computational Methods for Privacy & Explainability Compliance*, STOR Colloquium, University of North Carolina (UNC), Chapel Hill
- 2024 ◇ *Machine Unlearning Fails to Remove Data Poisoning Attacks*, Spotlight Presentation 🏆, GenLaw Workshop, ICML 2024, Vienna
- ◇ *Unlearning Private Data from Large Language Models: Algorithms and Evaluations*, DSE Lab, hosted by Prof. Jiliang Tang, Michigan State University
- ◇ *Guarding Private Data from Membership Inference Adversaries*, Privacy Workshop, TU Delft, Netherlands
- ◇ *On the Privacy Risks of Counterfactual Explanations for Algorithmic Recourse*, Workshop on Interpretability and Algorithmic Recourse, Saarland University, Germany
- 2023 ◇ *On the Tradeoffs between Actionable Explanations and the Right to be Forgotten*, Max-Planck Institute for Privacy & Security, Bochum, Germany
- ◇ *On the Tradeoffs between Actionable Explanations and the Right to be Forgotten*, Statistical Data Science Seminar, LMU Munich
- ◇ *On the Privacy Risks of Counterfactual Explanations for Algorithmic Recourse*, TUM Reliable Data Science Seminar, Technical University of Munich, Germany
- ◇ *On the Privacy Risks of Counterfactual Explanations for Algorithmic Recourse*, 2. XAI Winter School, Nice, France
- 2022 ◇ *Lecture on Realistic & Robust Counterfactual Explanations for Algorithmic Recourse*, XAI Summer School, TU Delft, Netherlands
- ◇ *On the Tradeoffs between Actionable Explanations and the Right to be Forgotten*, 1. XAI Winter School, Nice, France
- 2021 ◇ *CARLA: A Python Library to Benchmark Counterfactual Explanation Algorithms*, Spotlight Presentation 🏆, Algorithmic Recourse Workshop, ICML 2021
- ◇ *On Realistic & Robust Counterfactual Explanations for Algorithmic Recourse*, AI4Life Lab, Harvard University, Remote
- ◇ *On Realistic & Robust Counterfactual Explanations for Algorithmic Recourse*, JP Morgan AI Research, London, UK

Teaching

- Winter 2018/2019 ◇ **Data Mining and Probabilistic Reasoning, Teaching assistant**
 - ◇ Held tutorials, designed homework assignments, gave guest lectures, held office hours, and graded homeworks and final exam.
 - ◇ 110 MSc students
- Summer 2019 ◇ **Trustworthy Machine Learning, Co-Lecturer**
 - ◇ Designed the course curriculum, gave lectures, held office hours, and graded student reports and code submissions.
 - ◇ 15 MSc students
- Winter 2019/2020 ◇ **Data Mining and Probabilistic Reasoning, Teaching assistant**
 - ◇ 120 MSc students
- Summer 2020 ◇ **Trustworthy Machine Learning, Co-Lecturer**
 - ◇ 30 MSc students
- 2018-2023 ◇ Supervised 2 research projects, 4 MSc theses
- Summer 2026 ◇ **Mining Massive Data, Lecturer**
 - ◇ Modernized course curriculum, held lectures and held office hours.

Research Mentoring

- 2026 ♦ Hubert Baniecki (University of Warsaw), Vincent Hanke (CISPA)
- 2025 ♦ Samira Abedini (Saarland University), Sina Mavali (CISPA)
- 2024 ♦ Lillian Sun, Catherine Huang (BSc students, all Harvard University)
- 2023 ♦ Jimmy Z. Di (MSc student, University of Waterloo; ICLR'25)
- 2022 ♦ Vadim Borisov, Kathrin Sessler, Tobias Leemann (PhD students, all University of Tübingen; ICLR'23, NeurIPS'23, AAAI'24)
- 2021 ♦ Teresa Datta (MSc student, Harvard University; ICLR'23)
 ♦ Johan van-den Heuvel (MSc student, University of Tübingen; ICLR'23 & NeurIPS'21)
- 2020 ♦ Sascha Bielawski (MSc student, University of Tübingen; NeurIPS'21)

Workshops and Tutorials

- 2025 ♦ **ICLR workshop**
 "Building Trust in LLMs and LLM Applications: From Guardrails to Explainability to Regulation"
 <https://building-trust-in-llms.github.io/iclr-workshop/>
 with Hima Lakkaraju, Micah Goldblum, Ramasuri Narayanam, Soumyabrata Pal and Bang An
- 2022 ♦ **Tutorial at XAI Summer School**
 CARLA: A Python Library to Benchmark Counterfactual Explanation Algorithms
 TU Delft, Netherlands

Reviewing and Service

- Area Chair ♦ 2026: ICLR, KDD, NeurIPS
- Reviewer ♦ 2026: SatML
- ♦ 2025: ICLR, ICML, NeurIPS, TMLR
- ♦ 2024: ICLR, ICML, NeurIPS (outstanding reviewer 🏆), TMLR
- ♦ 2023: AISTATS, FAccT, ICML, JMLR, NeurIPS & NeurIPS DSB Track, ICLR
- ♦ 2022: AISTATS, AAAI, NeurIPS (outstanding reviewer 🏆) & NeurIPS DSB Track, ICML, FAccT
- ♦ 2021: NeurIPS & NeurIPS DSB Track, AAAI

References

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Google & Assistant Professor of Computer Science
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